Prism - Supporting Research Summary

Research Objective

Validate whether the core concept behind Prism exists today, understand adjacent solutions, explore academic foundations for flexible information structures, and assess whether AI enables new approaches to knowledge exploration.

Key Findings

- The opportunity is not better search — it is better exploration.
- Research supports the idea that information should not be limited to a single structure.
- AI creates a new possibility: dynamically generated knowledge structures.
- The opportunity is helping people navigate uncertainty.

Competitive Analysis

Product / Category	Core Experience	Strength	Gap / Opportunity for Prism
ChatGPT / Claude / Gemini AI Assistants	Conversational question answering and content generation	Natural language interaction; can explain complex topics	Primarily optimized for answering a question rather than helping users explore multiple ways of understanding a topic
Perplexity / AI Search AI-powered search	Search + synthesized answers with sources	Faster research and information retrieval	Still follows a search paradigm: users need to formulate questions and guide exploration

Google Search Traditional search	Keyword-based retrieval	Best-in-class information discovery	Requires users to know what they are looking for; does not reorganize information around user intent
Wikipedia Knowledge repository	Structured knowledge exploration	Rich interconnected information	Uses predefined categories and article structures
Amazon / Commerce Platforms Product discovery	Search, filtering, recommendations	Helps users narrow options	Categories and filters are designed around inventory and business needs, not individual goals
Notion / Obsidian / Roam Personal knowledge management	User-created information systems	Flexible storage and relationships between ideas	Users must manually create structure, tags, and connections
Knowledge Graph Platforms TheBrain, Kumu	Visualize relationships between concepts	Shows connections beyond folders	Typically requires human-curated relationships and does not dynamically adapt to intent

Insight - Competitive Analysis

Most existing solutions fall into three categories:

- **Retrieval**
  - *Goal:* Help users find an answer.
  - *Examples:* Google Search, Perplexity, ChatGPT

    - *Limitation:* Users must know what question to ask and guide the exploration.
- **Organization**
    - *Goal:* Help users store and structure information.
    - *Examples:* Notion, Obsidian, Roam
    - *Limitation:* Users are responsible for creating categories, tags, and relationships.
- **Visualization**
    - *Goal:* Help users see connections between concepts.
    - *Examples:* Knowledge graphs, mind maps, tools like TheBrain and Kumu
    - *Limitation:* Relationships are often manually created and do not adapt to the user's intent.

Prism explores a fourth category:

- **Adaptive Exploration**
    - *Goal:* Help users understand information through the perspective most useful for their current goal.
    - *Opportunity:* Instead of forcing users to adapt to a predefined structure, information adapts to the way they think and what they are trying to accomplish.

Existing Academic Research

Concept	What It Means	Relevance to Prism
Faceted Classification	Organizing information through multiple independent dimensions rather than a single hierarchy.	Supports the idea that the same information can be explored through different lenses depending on user goals.
Ontologies	Defining concepts and the relationships between them.	Enables richer understanding by connecting ideas beyond simple categories.

Knowledge Graphs	Representing information as interconnected entities and relationships rather than folders or lists.	Provides the foundation for exploring information through relationships and connections.
Sensemaking	The human process of creating understanding by identifying patterns, relationships, and meaning within complex information.	Aligns with Prism's goal of helping users navigate unfamiliar topics.
Multiple Classification Systems	Recognizing that different classification schemes can be valid depending on purpose and context.	Supports the core principle that there is not one "correct" way to organize information.

Insight - Academic Research

Traditional information systems often require a **single organizational structure**, but research shows that people benefit from **multiple ways of structuring and exploring knowledge**.

AI Categorization Capabilities

AI Capability	What It Enables	Relevance to Prism
Entity extraction	Identifying key concepts, objects, and topics from unstructured information	Allows AI to understand the building blocks of a topic

Relationship extraction	Identifying connections between concepts	Enables exploration beyond traditional categories
Ontology generation	Creating structured representations of concepts and relationships	Supports dynamic knowledge models
Taxonomy generation	Creating hierarchical or multi-dimensional groupings	Enables alternative ways to organize the same information
Knowledge graph construction	Representing information as connected entities and relationships	Provides the foundation for relationship-based exploration

Insight - AI Capabilities

- Traditional approach:
  - Human experts define categories → Software organizes information
- AI-enabled approach:
  - Information → AI identifies concepts and relationships → Users explore through relevant perspectives
- This shift creates the possibility of **adaptive information experiences** where the structure changes based on the user's goal rather than remaining fixed.

Resources & References

Competitive Landscape

- ChatGPT — AI assistant and conversational knowledge exploration
  https://chat.openai.com/
- Perplexity — AI-powered search and research
  https://www.perplexity.ai/
- Notion — Personal knowledge management and organization
  https://www.notion.com/

- Obsidian — Connected notes and personal knowledge graphs
  https://obsidian.md/
- TheBrain — Visual knowledge graph and relationship mapping
  https://thebrain.com/
- Kumu — Relationship mapping and systems visualization
  https://kumu.io/

Information Science & Knowledge Organization

- Faceted Classification — Research on organizing information through multiple dimensions rather than a single hierarchy
  https://www.sciencedirect.com/topics/computer-science/faceted-classification
- Microsoft Research: Faceted Search — Research on helping users explore information through multiple dimensions
  https://www.microsoft.com/en-us/research/project/faceted-search/
- Semantic Web Journal: Large Language Models for Ontology Engineering — Research on using LLMs to create and manage structured knowledge models

  https://www.semantic-web-journal.org/content/large-language-models-ontology-engineering-systematic-literature-review-1
- Google Knowledge Graph — Overview of representing information through entities and relationships
  https://blog.google/products/search/introducing-knowledge-graph-things-not/
- Sensemaking — Research area focused on how humans create understanding from complex information
  https://en.wikipedia.org/wiki/Sensemaking

AI & Dynamic Knowledge Structures

- Large Language Models and Knowledge Graphs: A Survey — Research on combining LLMs with structured knowledge representations
  https://arxiv.org/abs/2406.08223
- Accelerating Knowledge Graph and Ontology Engineering with Large Language Models — Research on AI-assisted knowledge modeling
  https://www.sciencedirect.com/science/article/pii/S1570826825000022